

Internship Final Report

Student Name: Chinmayi S

University: VTU

Major: Include your Major

Internship Duration: September-1st, 2024 - September 30th, 2025

Company: ShadowFox

Domain: AIML

Mentor: Mr. Hariharan

Coordinator: Mr. Aakash

Objectives

The primary objective of this internship project was to build an **Image Classification (Image Tagging) system** using a **Convolutional Neural Network (CNN)** trained on the **CIFAR-10 dataset**.

The key goals were:

- To understand and implement the end-to-end machine learning pipeline: data preprocessing, model building, training, evaluation, and deployment.
- To gain hands-on experience with **TensorFlow and Keras** for deep learning applications.
- To develop documentation and proof-of-work (GitHub repository, report, and LinkedIn post) as part of professional project submission.

Tasks and Responsibilities

During this project, the following tasks were carried out:

- **Data Preprocessing:** Normalized CIFAR-10 images, applied data augmentation (random flips, rotations, zoom) to improve generalization.
- **Exploratory Data Analysis (EDA):** Visualized sample images and verified dataset structure to understand the problem domain.
- **Model Development:**
 - Built a CNN with multiple convolutional and pooling layers.
 - Added dropout layers to reduce overfitting.
 - Used Adam optimizer and categorical cross-entropy loss.

- **Model Training:** Trained the model on the training dataset with early stopping and model checkpointing to save the best version.
- **Evaluation:** Assessed performance using accuracy and loss metrics on the test dataset.
- **Visualization:** Plotted training curves (loss and accuracy) and displayed sample predictions (true vs. predicted labels).
- **Documentation & Proof of Work:**
 - Created a GitHub repository with README, notebooks, saved models, and reports.
 - Took screenshots of results and saved proof of work.
 - Recorded a short video and posted it on LinkedIn tagging ShadowFox.

Learning Outcomes

This project helped me achieve the following learning outcomes:

- Gained practical understanding of **Convolutional Neural Networks (CNNs)** for image classification tasks.
- Learned to use **TensorFlow/Keras** for model building, training, and saving models in .h5 and .keras formats.
- Understood the importance of **data preprocessing and augmentation** to improve model generalization.
- Improved skills in **model evaluation** (accuracy, loss curves, prediction visualization).
- Learned how to structure a project professionally using GitHub (repo structure, README, reports, models, screenshots).
- Strengthened communication skills by preparing a **report, LinkedIn proof-of-work post, and a short demo video**.
- Experienced the end-to-end cycle of a machine learning project, from dataset handling to deployment-ready outputs.

Challenges and Solutions

- **Overfitting:** The model started memorizing training data and performed poorly on validation.
- **Solution:** Introduced dropout layers, applied data augmentation, and used early stopping to prevent over-training.
- **Training Time:** Training deep CNNs on CPU was slow.
- **Solution:** Enabled GPU runtime in Google Colab, which significantly accelerated training.
- **Saving Models:** Faced errors while saving models without proper file extensions.
- **Solution:** Used .keras and .h5 extensions as recommended in TensorFlow's new saving API.
- **Accuracy Limitations:** CIFAR-10 is a challenging dataset with small image sizes (32×32).
- **Solution:** Tuned hyperparameters (batch size, epochs), experimented with augmentation, and added dropout to stabilize performance.

Conclusion

This internship project was a valuable learning experience that allowed me to apply deep learning techniques in a practical setting. Key achievements include:

- Built and trained a CNN achieving a decent test accuracy (~70–75%) on CIFAR-10.
- Successfully implemented data preprocessing, augmentation, model training, and evaluation in a structured pipeline.
- Documented the project thoroughly with GitHub repo, final report, screenshots, and LinkedIn proof-of-work.

The project gave me exposure to the end-to-end AI/ML workflow and prepared me for handling more advanced projects in the future. It strengthened both my technical and professional skills, aligning well with the objectives of the ShadowFox AI/ML internship.

Acknowledgments

I sincerely thank **ShadowFox** for providing me with this internship opportunity. The structured tasks and proof-of-work approach helped me improve my technical knowledge, project management skills, and confidence in presenting my work publicly.